CS & IT ENGINEERING

Operating System

Memory Management

Lecture - 4

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Recap of Previous Lecture

Topic Non-Contiguous Memory Mgmt. Technique

Topic Paging

Topics to be Covered

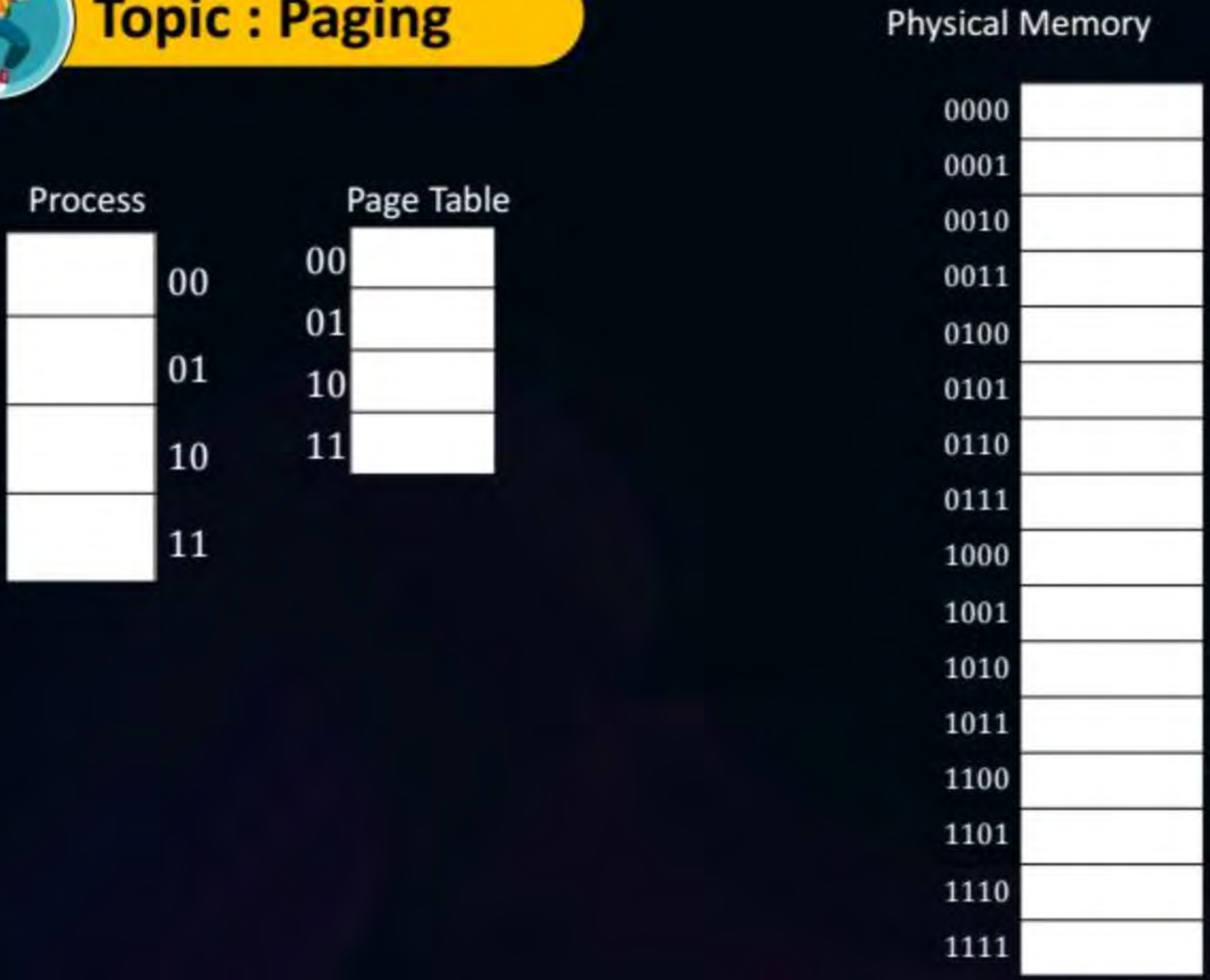
- **Topic** Paging
- **Topic** TLB

Topic : Paging

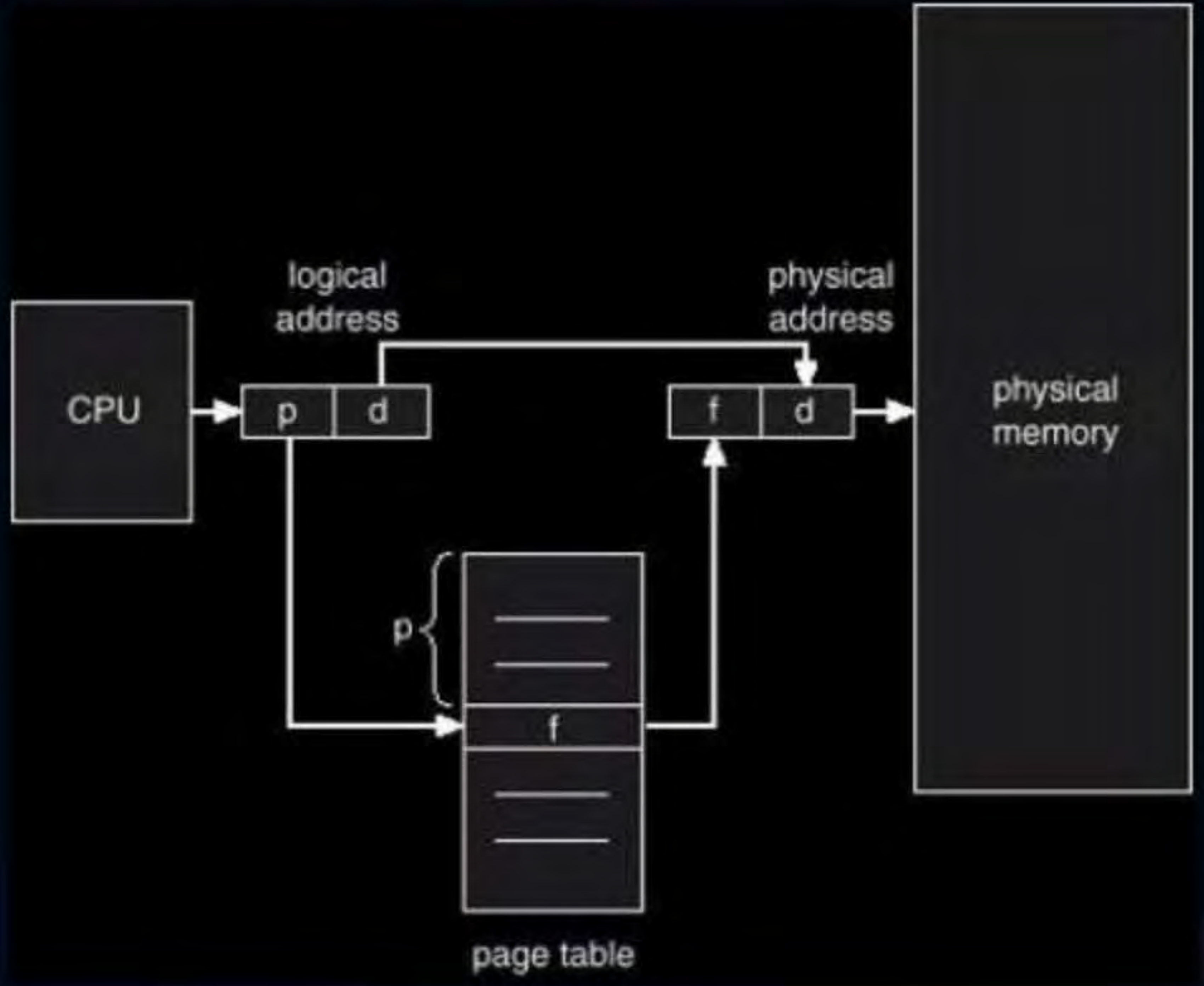
Consider

- A process has 4 pages
- Main memory has 8 frames

Topic : Paging



Topic : Paging



Page table size = no. of pages * 1 Page table entry size

1 P.T.E. size = (frame no. bits + extra bits)

Ques) L.A. space = 16k bytes $= 2^4 \cdot 2^{10}$ B $= 2^{14}$ B $\Rightarrow$ L.A. = 14 bits

P.A. space = 1 M bytes $= 2^{20}$ B $\Rightarrow$ P.A. = 20 bits

Page size = 128 bytes $= 2^7$ B $\Rightarrow$ offset (d) = 7 bits

Method 1 :-

no. of pages $= \dfrac{16K\cancel{B}}{128\cancel{B}} = \dfrac{2^{14}}{2^7} = 2^7 \Rightarrow$ Page no. = 7 bits

no. of frames $= \dfrac{1M\cancel{B}}{128\cancel{B}} = \dfrac{2^{20}}{2^7} = 2^{13} \Rightarrow$ frame no. = 13-bits

Page size = 128 B $= 2^7$ B $\Rightarrow$ d = 7 bits

$\xleftarrow{\quad}$ L.A. 14 $\xrightarrow{\quad}$

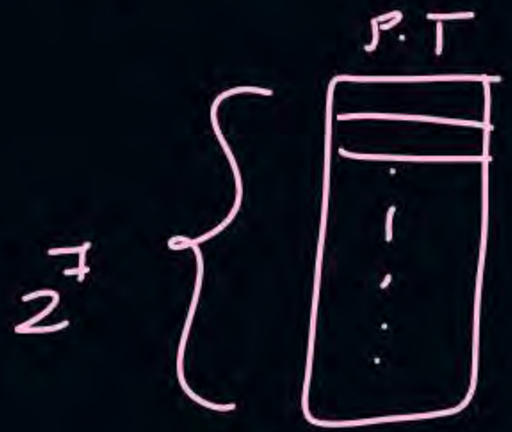
p	d
7	7

P.A.

$\xleftarrow{\quad}$ 20 $\xrightarrow{\quad}$

f	d
13	7

Page table size = No. of pages * 1 Page table entry (P.T.E.) size



$= 2^7 * 13$ bits

Method 2 :-

L.A. ← 14 →

p	d
7	7

⇓

no. of pages = 2^7

P.A. ← 20 →

f	d
13	7

⇓

no. of frames = 2^{13}

Topic : Question

#Q. Consider a paged memory system where the logical address of 28 bits and physical address of 34 bits. If each page table entry is of 4 bytes and page size is 2KB then:

$4 * 8$ bits $= 32$ bits

1. Number of pages in process? 2^{17}
2. Number of frames in main memory? 2^{23}
3. Number of bits for page number? 17 bits
4. Number of bits for frames? 23 bits
5. Number of entries in page table? 2^{17}
6. Page table size? $2^{17} * 4B = 2^{17} * 2^{2} B = 2^{19} B = \boxed{512 \text{ KBytes}} = 512K * 8 \text{ bits} = 2^{22} \text{ bits} \boxed{= 4M \text{ bits}}$

L.A.

$\longleftarrow 28 \longrightarrow$

p	d
17	11

P.A.

$\longleftarrow 34 \longrightarrow$

f	d
23	11

Page size = 2 kbytes = $2^1 \cdot 2^{10}$ bytes = 2^{11} bytes $\Rightarrow d = 11$ bits

---

no. of extra bits per P.T.E. = 32 − 23 = 9 bits

P.T. entry

frame no.	extra bits

Topic : Question

[GATE-2001]

$2^{26}\,B \Rightarrow P.A. = 26$ bits

#Q. Consider a machine with 64 MB physical memory and a 32-bit ~~virtual~~ logical address ~~space~~. If the page size is 4 KB, what is the approximate size of the page table?

$\rightarrow 2^{2}\cdot 2^{10}\,B = 2^{12}\,B \Rightarrow d = 12$ bits

A. 16 MB

B. 8 MB

C. ✓ 2 MB

D. 24 MB

L.A.

$\longleftarrow 32 \longrightarrow$

p	d
20	12

P.A.

$\longleftarrow 26 \longrightarrow$

f	d
14	12

$$P.T.\ \text{Size} = 2^{20} * (14)\ \text{bits}$$

$$= 14\ \text{M bits} \cong 2\,\text{MB}$$

[MCQ]

#Q. Consider a paged memory system where the process size is 16MB ($2^{24}B$) and main memory size is 4GB ($2^{32}B$). The page size is 2KB. $\rightarrow 2^{11}B$

- **A** Number of pages in process? 2^{13}
- **B** Number of frames in main memory? 2^{21}
- **C** Number of bits for page number? 13
- **D** Number of bits for frames? 21
- **E** Number of entries in page table? 2^{13}
- **F** Page table size? $\rightarrow 2^{13} * 21$ bits

L.A.

$\longleftarrow 24 \longrightarrow$

p	d
13	11

P.A.

$\longleftarrow 32 \longrightarrow$

f	d
21	11

[MCQ]

#Q. Consider a paged memory system where the process size is 128MB and main memory size is 2GB. The page size is 1KB.

2^{31} 2^{27} $\rightarrow 2^{10}$

A Number of pages in process? 2^{17}

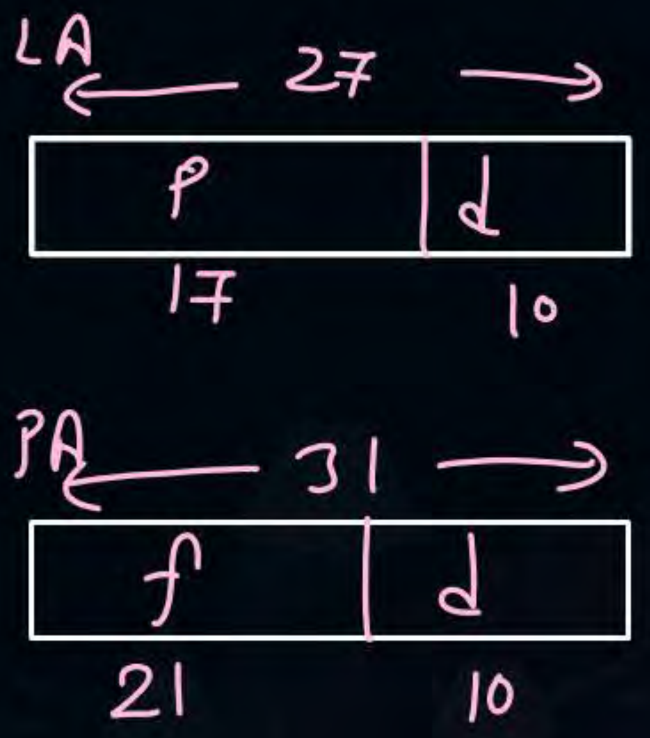
B Number of frames in main memory? 2^{21}

C Number of bits for page number? 17

D Number of bits for frames? 21

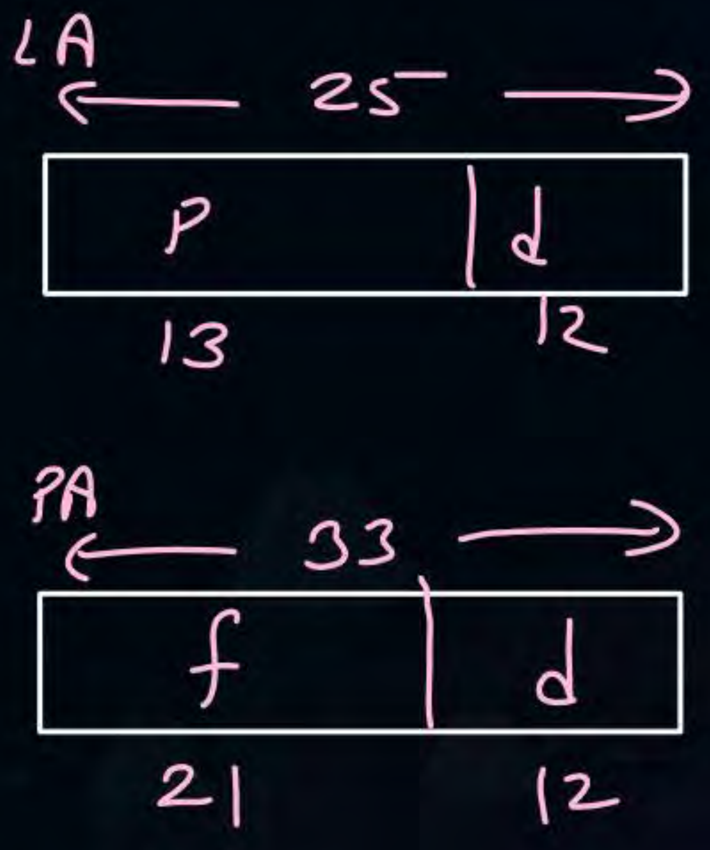
E Number of entries in page table? 2^{17}

F Page table size? $2^{17} * 21$ bits



[MCQ]

#Q. Consider a paged memory system where the logical address is 25 bits and physical address is 33 bits. The page size is 4KB. $\rightarrow 2^{12}$ B



A Number of pages in process? 2^{13}

B Number of frames in main memory? 2^{21}

C Number of bits for page number? 13

D Number of bits for frames? 21

E Number of entries in page table? 2^{13}

F Page table size? $2^{13} * 21$ bits

[MCQ]

[GATE-2015]

#Q. A computer system implements 8 kilobyte pages and a 32–bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the logical address supported by the system is <u>36</u> bits?

frame no.

Page size $= 8kB = 2^{13}B$

P.A. = 32 bits

P.T.E. = 1 valid bit + 1 dirty bit + 3 permission bits + frame no.

Page Table size = 24 Megabytes

L.A.

p	d
23	13

$\underbrace{23 \quad 13}_{36 \text{ bits}}$

P.A.

$\xleftrightarrow{32}$

f	d
19	13

P.T. size = no. of pages * 1 P.T.E. size

24M bytes = no. of pages * 24 bits

$\cancel{24}M * 8\ \cancel{bits}$ = no. of pages * $\cancel{24}\ \cancel{bits}$

no. of pages = 8M

$= 2^{23}$

$\Downarrow$

page no. = 23 bits

1 P.T.E. size = (19 + 1 + 1 + 3) bits

= 24 bits

Topic : Paging

Where the Page table Stored?

⇓

in main memory

Topic : Paging

Performance of Paging

Effective mem. access time $= \underline{2 * t_{mm}}$

(on avg. mem access time)

one for page table

one for content

where, t_{mm} = main mem. access time

If page table is small and if stored in Register.

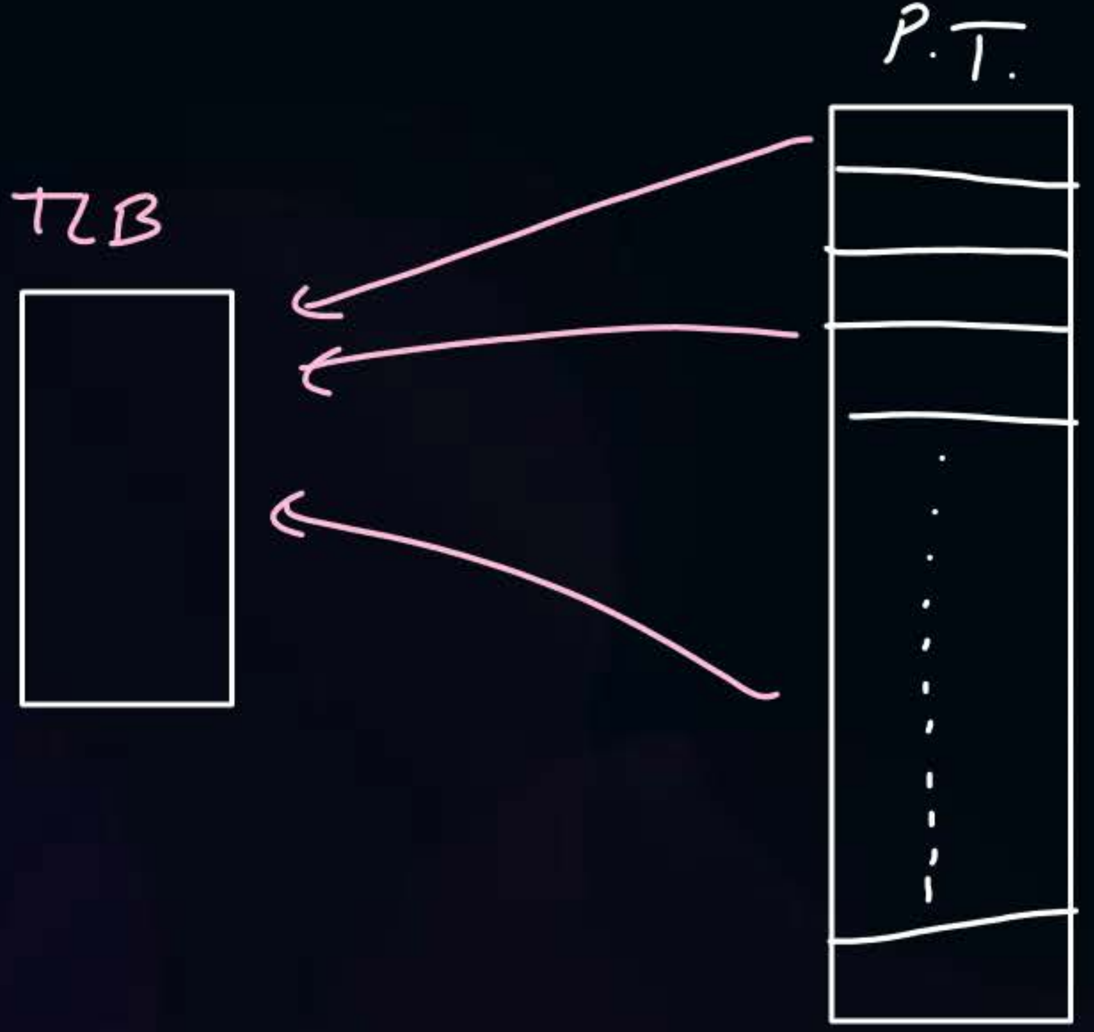
$$E.M.A.T. = t_{mm}$$

(P.T. access is done from register that takes negligible time

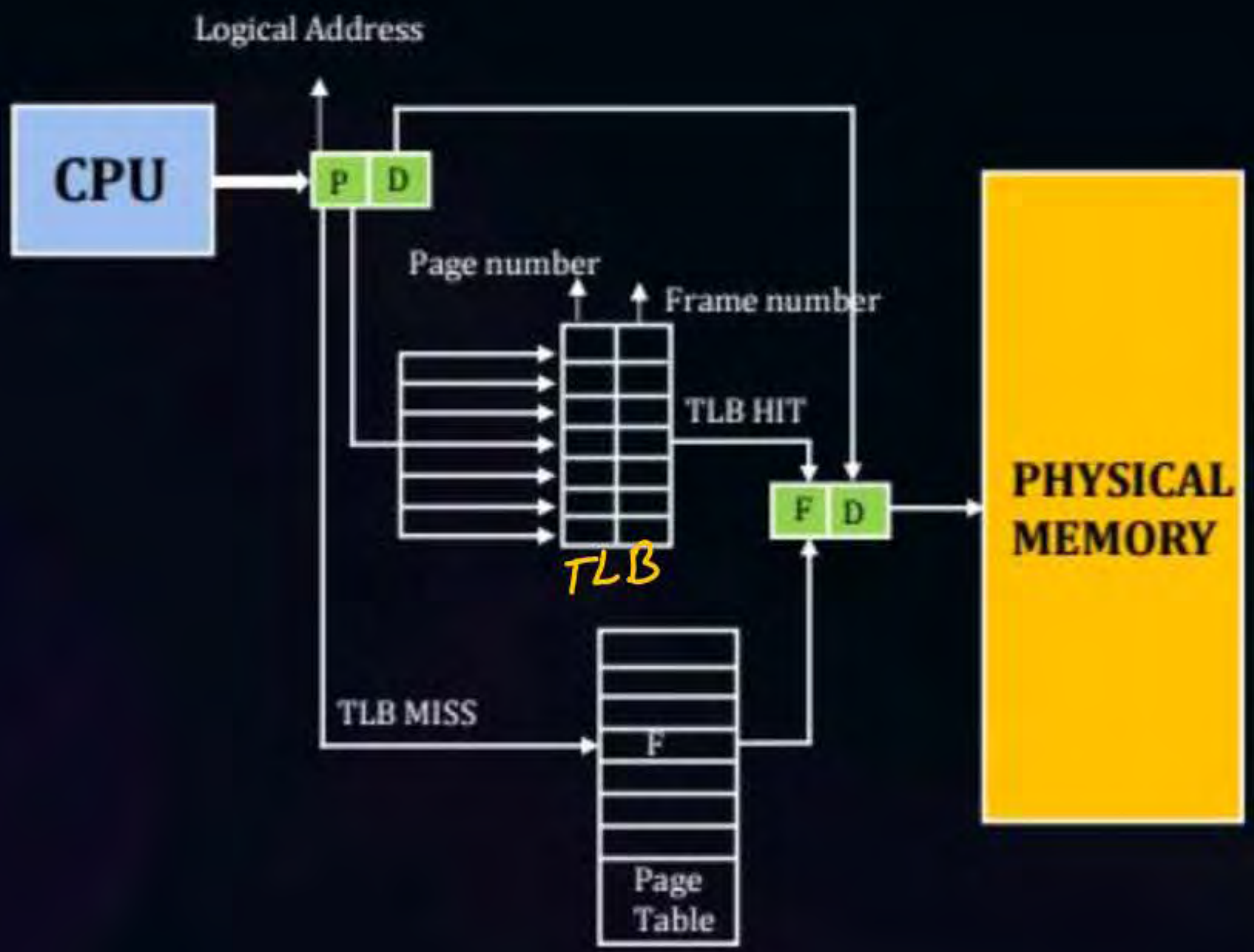
Topic : Paging

TLB (Translation Lookaside Buffer) → used to store a few most frequently & recently used P.T. entries.

(smaller & faster Hardware)



TLB (Translation Lookaside Buffer)



H = hit ratio of TLB | t_{TLB} = TLB access time

$$E.M.A.T. \text{ with TLB} = H * (\underbrace{t_{TLB}}_{\text{translation}} + \underbrace{t_{mm}}_{\text{for content}}) + (1-H)(\underbrace{t_{TLB}}_{\text{searching in TLB}} + \underbrace{2t_{mm}}_{\text{one for PT, one for content}})$$

or

$$= t_{TLB} + t_{mm} + (1-H)\, t_{mm}$$

ex :-

$t_{TLB} = 15\ ns$

$t_{mm} = 300\ ns$

$H = 80\%$

E.M.A.T. without TLB $= 2 * 300 = 600\ ns$

E.M.A.T. with TLB $= 15 + 300 + 0.2 * 300$

$= 375\ ns$

speed up or

performance gain of using TLB $= \dfrac{\text{slower technique time}}{\text{faster technique time}}$

$$= \frac{600\,ns}{375\,ns}$$

$$= 1.6$$

2 mins Summary

Topic Paging

Topic TLB

Happy Learning

THANK - YOU